

ON-DEVICE AI

YOUR PERSONAL SMART ASSISTANT

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yped up technological phenomena are quite common and most of them end up being a bubble. One that pops without mercy and takes out most of the companies in that domain. The surviving few are the ones who've truly innovated or offer a little more than just a buzzword. AI is no different. There are several key players whose names we all know for they've become synonymous with AI, for eg. ChatGPT. It's a Large Language Model (LLM) based chatbot and the most popular of all AI services currently available to the public. And there are many like it but most people will only recall ChatGPT.

ChatGPT basically consists of a weighted AI model that's running on a server somewhere with access to petabytes of information which it can access at the drop of a hat. For anyone who's used ChatGPT, they'll be all too familiar with the time taken for each response. That's because it takes time for the model to run through all the information that it has and generate an adequate response. And the reason why it can give you a response as fast as it currently does is because the model is running on some of the most powerful hardware that money can buy. So does that mean that AI will forever remain on the cloud? Not quite.

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AI requires a lot of horsepower for the training stage i.e. when it is learning. Once deemed to be worthy of being put to use, the AI model no longer requires powerful hardware, it can make do with a fraction of the processing power. This stage is termed as inference. So once an AI is trained, it can function on a relatively low-power processor. You can opt for a more energy efficient application by using a processor that is designed for AI applications. Say, something like your average smartphone. And if you haven't realised, most smartphones have already incorporated some form of AI processor for a few generations. A lot of companies call this specialised processor as the Neural Processing Unit (NPU). And unlike the cloud-reliant AI systems of the past, on-device AI brings the power of artificial intelligence directly into the hands of users, offering a blend of personalisation, security, and efficiency previously unimaginable. Previous iterations of the NPU have been quite subtle with very small AI models running off of it for purposes of polishing up that photo that you just shot with your phone camera. With the advent of the AI arms race, all smartphone hardware manufacturers have ramped up efforts significantly allowing for much more

capable models to operate directly on your personal device.

Tracing back the lineage of on-device AI, we uncover a history rich with innovation and breakthroughs. From the initial, simple algorithms to today's sophisticated models capable of running directly on devices, the journey has been nothing short of remarkable. Take for example the recent announcements by Samsung and Google. Their new devices can actually translate real-time phone calls in completely different languages, ramp up image and video quality, transcribe conversation and generate minutes of the meeting and manipulate photos and videos like a movie studio from the '80s. I say '80s because there are a lot of funny artefacts produced when you try out these features. While Google has its own arm-based processor for its Pixel devices the other device manufacturers have to rely on the industry giants to bring the power of AI on their devices. Qualcomm, a frontrunner in this revolution, has been instrumental in pushing the boundaries of what's possible, pioneering the development of sub-10 billion parameter models that bring generative AI to devices small and large without compromising on performance. This evolution underscores a significant trend: the movement towards making AI ubiquitous, yet unobtrusive, in our daily lives.

Today, on-device AI is not confined to the realm of smartphones and tablets,